

Computer Networks: TCP/IP Protocol Suite

Chapter 1: Introduction to TCP

TCP (Transmission Control Protocol) is a core protocol of the Internet Protocol Suite. TCP provides reliable, ordered, and error-checked delivery of a stream of octets (bytes) between applications running on hosts communicating via an IP network.

TCP is connection-oriented, meaning that a connection between client and server is established before data can be sent. The connection is established using the three-way handshake: SYN, SYN-ACK, ACK.

Key Features of TCP:

- Reliable delivery with acknowledgment
- Flow control using sliding window protocol
- Error detection using checksums
- Congestion control algorithms
- Full-duplex communication

TCP Congestion Control

TCP Congestion Control is one of the most important features of TCP. It prevents any single network connection from using too much bandwidth.

Slow Start Algorithm:

When a new connection is established, TCP starts with a small congestion window (cwnd) and increases it exponentially until a threshold is reached.

This algorithm is called slow start because it starts with a small cwnd.

Congestion Avoidance:

After reaching the slow start threshold (ssthresh), TCP switches to congestion avoidance where cwnd grows linearly (additive increase).

Fast Retransmit and Fast Recovery:

When 3 duplicate ACKs are received, TCP retransmits the lost packet without waiting for a retransmission timeout.

TCP then uses fast recovery to reduce cwnd by half (multiplicative decrease).

Modern congestion control algorithms include: CUBIC, BBR (Bottleneck Bandwidth and Round-trip propagation time), and QUIC.

IP Protocol and Routing

The Internet Protocol (IP) is the principal communications protocol for relaying datagrams across network boundaries.

IPv4 Addressing:

IPv4 uses 32-bit addresses, providing approximately 4.3 billion unique addresses.

Addresses are written in dotted-decimal notation: 192.168.1.1

IPv6 Addressing:

IPv6 uses 128-bit addresses to overcome IPv4 address exhaustion.

Addresses are written as eight groups of four hexadecimal digits.

Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Routing Protocols:

- OSPF (Open Shortest Path First) - link state routing
- BGP (Border Gateway Protocol) - path vector routing for the internet
- RIP (Routing Information Protocol) - distance vector routing

Network Address Translation (NAT):

NAT allows multiple devices on a private network to share a single public IP address.